

International Islamic University Chittagong (IIUC)

Department of Science of Hadith and Islamic Studies

Mid-Term Examination, Spring-2026

Course Code: BSE-2301

Full marks: 30

Course Title: Basic Sciences

Time: 1 hour and 30 minutes

Bloom's Levels of the questions						
Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember

Questions (Answer any three of the following questions)	Marks	Bloom's Level	CLOs
1. a) Convert 0.375 into a common fraction.	2	Understand	CLO1
1. b) Simplify $\frac{5 * 3^n - 2 * 3^{n-1}}{3^n}$	2	Apply	CLO2
1. c) Find the sum: 7+10+13+...+70	2	Analyze	CLO3
1. d) Factorize $x^2 - 9x + 20$	2	Evaluate	CLO2
1. e) Find 18% of 250.	2	Remember	CLO1
2. a) The difference of two numbers is 60. Also, 20% of the first number equals 30% of the second number. Find the numbers.	3	Create	CLO3
2. b) Solve $3^{x+2} = 9^{x-1}$	2	Apply	CLO1
2. c) The ratio of boys to girls is 3:4 and the ratio of girls to teachers is 4:1. If the total number is 96, find the number of boys, girls, and teachers.	5	Evaluate	CLO2
3. a) The area of a triangle is 84 cm ² and its base is 12 cm. Find its height.	3	Apply	CLO3
3. b) The length of a rectangle is 4 cm more than its width. If the perimeter is 40 cm, find its area.	4	Analyze	CLO3
3. c) Factorize $x^2 + 7x + 10$	3	Apply	CLO2
4. a) Write in decimal form: - Seven hundred twenty-five thousandths $\frac{13}{8}$ (as decimal)	2	Remember	CLO1
4. b) Find the missing value: $\frac{4}{9} = \frac{x}{45}, \frac{7}{8} = \frac{56}{x}$	2	Analyze	CLO3
4. c) Arrange in descending order: $\frac{5}{6}, \frac{7}{12}, \frac{3}{4}$	2	Understand	CLO2
4. d) A circle has an area of 154 cm ² . Find its diameter.	2	Apply	CLO3
4. e) Divide $\frac{5}{12} \div 2\frac{1}{3}$	2	Understand	CLO1

International Islamic University Chittagong (IIUC)

Faculty of Shariah and Islamic Studies (FSIS)

Dept. of Science of Hadith & Islamic Studies (SHIS)

Mid-Term Examination, Autumn-2025

Course Code: BSE-2301

Course Title: Basic Sciences

Full marks: 30

Time: 1 hour 30 minutes

Instructions : (i) Programmable calculators are not allowed. Do not write anything on the question paper.
(ii) Answer any Three of the following questions. (iii) Marks allocated are indicated in the right margin.

Bloom's Levels of the questions						
Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember

Serial No	Questions	Marks	Bloom's level	CLOs
1.	a Convert $42.347\bar{8}$ into common fraction.	2	Understand	COL-1
	b Simplify $\frac{3.2^n - 4.2^{n-2}}{2^n - 2^{n-1}}$	2	Apply	COL-2
	c $5+11+17+23+\dots+59 = \text{what?}$	2	Analyze	COL-3
	d Find the factors of $x^2+10x+25$	2	Evaluate	
	e What is 15% of 200?	2	Understand	COL-1
2.	a Difference of two numbers X and Y, ($X>Y$) is 100. Also 10% of X is equal to 15% Y. Find the two numbers.	3	Create	COL-3
	b If $2^{x+7} = 4^{x+2}$; then $x=?$	2	Apply	COL-1
	c The ratio of boys to girls in Class Nine is 4 : 5, and the ratio of girls to teachers is 5 : 1. If the total number of boys, girls, and teachers together is 110, find the number of boys, girls, and teachers.	5	Evaluate	COL-2
3.	a When the length of each side of a triangle is increased by 2 meter, its area is increased $6\sqrt{3}$ square meter. Find the length of side the triangle.	4	Analyze	COL-2
	b The length of a rectangle is 5 cm more than its width. If the perimeter is 36 cm, find the area of the rectangle.	4	Apply	COL-3
	c Find the square root of 625 using the division method.	2	Apply	COL-2
4.	a Write each decimal fraction. i. Six thousand twenty-seven and one hundred four millionths. ii. 2.00123	2	Remember	COL-1
	b Determine the missing numerator or denominator i. $\frac{3}{7} = \frac{?}{35}$ ii. $\frac{5}{6} = \frac{45}{?}$	2	Analyze	COL-3
	c Arrange $\frac{7}{12}, \frac{9}{24}, \frac{3}{8}$ in descending order (highest to lowest).	2	Understand	COL-2
	d A circle has an area of 154 cm^2 . Find its radius.	2	Apply	COL-3
	e Divide $2\frac{2}{9}$ by $3\frac{1}{3}$.	2	Understand	COL-1

International Islamic University Chittagong (IIUC)

Faculty of Shariah and Islamic Studies (FSIS)

Mid-Term Examination, Spring-2025

Course Code: BSE-2301

Full marks: 30

Course Title: Basic Sciences

Time: 1 hour 30 minutes

Answer any **Three** of the following questions.

Each question has **Ten** marks.

Bloom's Levels of the questions						
Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember

Serial No	Questions	Marks	Bloom's level	CLOs
1	a Find the square root of 225 using the division method	2	Understand	COL-1
	b Multiply $(3x + 2y)$ with $(4x + 6y - 8z)$.	2	Apply	COL-2
	c Simplify $4^{1.5}$.	2	Analyze	COL-1
	d Find the factors of $2x^2 + 7x - 15$.	2	Evaluation	
	e Find 20% of 120	2	Create	COL-1
2	a Perform the following operations: I. 246.135×125.37 II. $0.122 \div 0.02$	3	Remember	COL-1
	b If $2^{x+7} = 4^{x+2}$; then $x = ?$	2	Apply	COL-1
	c The ratio of DIS to QGIS new admission students is 5:2 and the ratio of QGIS to SHIS is 2:3. If the total number of students in the faculty of shariah and Islamic studies is 120, how many QGIS, DIS and SHIS students are there?	5	Understand	COL-2
3	a A triangle has an area of 60 cm^2 and a base of 15 cm. Find its height	4	Apply	COL-2
	b A rectangle has a length that is twice its width. if the perimeter is 60 cm, find its area.	4	Apply	COL-2
	c Factorize $x^2 + 5x + 6$	2	Understand	COL-2
4	a Write the following fractions using whole numbers. i. Eight hundred seven-thousandths ii. Three eighths	2	Remember	COL-1
	b Convert improper fraction to its corresponding mixed number $\frac{83}{11}$	2	Analyze	COL-1
	c Arrange $\frac{5}{12}, \frac{11}{24}, \frac{3}{8}$ in descending order (highest to lowest).	2	Understand	COL-1
	d Determine the missing denominator $\frac{5}{6} = \frac{45}{?}$	2	Apply	COL-1
	e Divide $2\frac{2}{9}$ by $3\frac{1}{3}$.	2	Create	COL-1